Math 54 — Week 10

The Gram–Schmidt process and QR; least squares; inner product spaces (Lay 6.4, 6.5, 6.7)

§1 Facts used from Weeks 1–9

(W1) **Projection formula.** If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal basis of a subspace W of $\mathbb{R}^n$, then

$$\operatorname{proj}_W \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p}\mathbf{u}_p,$$

and $\mathbf{y} - \operatorname{proj}_W \mathbf{y}$ is in $W^\perp$. The decomposition of $\mathbf{y}$ as (vector in W) + (vector in $W^\perp$) is unique (orthogonal decomposition theorem, Week 9, Thm. 8).

(W2) **Best approximation.** With W as in (W1), $\operatorname{proj}_W \mathbf{y}$ is the point of W closest to $\mathbf{y}$, and the only one: $\|\mathbf{y} - \operatorname{proj}_W \mathbf{y}\| < \|\mathbf{y} - \mathbf{v}\|$ for every other $\mathbf{v} \in W$ (Week 9, Thm. 9).

(W3) **Orthogonal sets and bases.** An orthogonal set of nonzero vectors is linearly independent (Week 9, Thm. 4); dividing each vector by its length gives an orthonormal set (Week 9). In a p-dimensional subspace, p independent vectors form a basis (Week 6).

(W4) **Orthonormal columns.** Q has orthonormal columns if and only if $Q^\mathsf{T}Q = I$ (Week 9, Thm. 6). A vector is orthogonal to $\operatorname{Col} A$ if and only if it is orthogonal to each column of A, that is, if and only if it is in $\operatorname{Nul} A^\mathsf{T}$ (Week 9, Thm. 3).

(W5) $\operatorname{Nul} A^\mathsf{T}A = \operatorname{Nul} A$ (Week 9, P6): if $A^\mathsf{T}A\mathbf{x} = \mathbf{0}$, then $\|A\mathbf{x}\|^2 = \mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x} = 0$, so $A\mathbf{x} = \mathbf{0}$; and if $A\mathbf{x} = \mathbf{0}$, then $A^\mathsf{T}A\mathbf{x} = \mathbf{0}$.

(W6) **Invertibility** (Weeks 4–5, Lay 2.3, 3.1–3.2). A square matrix is invertible if and only if its null space is $\{\mathbf{0}\}$, if and only if its determinant is not 0. $\operatorname{Nul} A = \{\mathbf{0}\}$ exactly when the columns of A are independent. The determinant of a triangular matrix is the product of its diagonal entries. If R is invertible, so is R^T, with $(R^\mathsf{T})^{-1} = (R^{-1})^\mathsf{T}$; and $(QR)^\mathsf{T} = R^\mathsf{T}Q^\mathsf{T}$.

(W7) **Pythagorean theorem** (Week 9). $\mathbf{u} \perp \mathbf{v}$ if and only if $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. Also $\|c\mathbf{v}\| = |c|\,\|\mathbf{v}\|$.

(W8) **Matrix–vector products** (Lay 1.4, 2.1, 2.2). $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$; entry (i, j) of $A^\mathsf{T}B$ is column i of A times column j of B. The inverse of a 2×2 matrix with rows (a, b), (c, d) and $ad - bc \neq 0$ has rows $(d, -b)$, $(-c, a)$ divided by $ad - bc$.

(W9) **Rank theorem** (Week 6). For an $m \times n$ matrix, $\operatorname{rank} A + \dim \operatorname{Nul} A = n$.

§2 This week

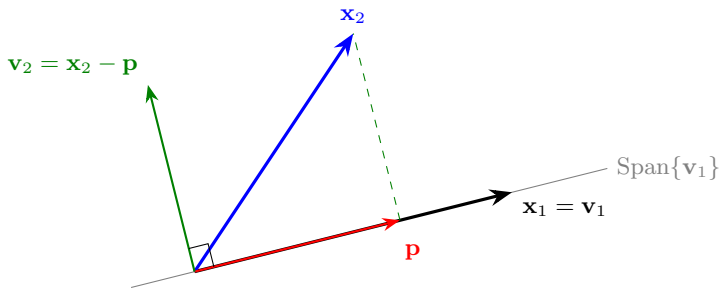
§2.1 The Gram–Schmidt process (Lay 6.4)

The question. The projection formula (W1) needs an *orthogonal* basis of W, but subspaces usually come with an ordinary basis. For $W = \operatorname{Span}\{(3, 6, 0), (1, 2, 2)\}$, whose two spanning vectors are not orthogonal ($3 + 12 + 0 = 15$), how do we find an orthogonal basis of W, and then of any subspace?

Two vectors. Let $W = \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$ with $\mathbf{x}_1, \mathbf{x}_2$ independent. Keep $\mathbf{v}_1 = \mathbf{x}_1$. From $\mathbf{x}_2$, subtract its projection onto the line through $\mathbf{v}_1$:

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1.$$

By (W1), $\mathbf{v}_2$ is orthogonal to $\mathbf{v}_1$. It lies in W because it is a combination of $\mathbf{x}_2$ and $\mathbf{x}_1$, and it is not $\mathbf{0}$ because $\mathbf{x}_2$ is not a multiple of $\mathbf{x}_1$. So $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis of W (W3).



One step of Gram–Schmidt: $\mathbf{p}$ is the projection of $\mathbf{x}_2$ onto the line through $\mathbf{v}_1$, and $\mathbf{v}_2 = \mathbf{x}_2 - \mathbf{p}$ is orthogonal to $\mathbf{v}_1$. The dashed segment from $\mathbf{p}$ to $\mathbf{x}_2$ is a copy of $\mathbf{v}_2$.

Example. $\mathbf{x}_1 = (3, 6, 0)$, $\mathbf{x}_2 = (1, 2, 2)$. Here $\mathbf{x}_2 \cdot \mathbf{v}_1 = 15$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 45$, so

$$\mathbf{v}_2 = (1, 2, 2) - \frac{15}{45}(3, 6, 0) = (0, 0, 2).$$

Check: $(3, 6, 0) \cdot (0, 0, 2) = 0$. So $(3, 6, 0)$, $(0, 0, 2)$ is an orthogonal basis of W.

For three or more vectors, repeat: subtract from each new $\mathbf{x}_k$ its projection onto the span of the vectors already built. That projection can be computed with (W1) because the vectors already built are orthogonal.

Theorem (The Gram–Schmidt process (Lay Thm. 11))

Let $\{\mathbf{x}_1, \ldots, \mathbf{x}_p\}$ be a basis of a nonzero subspace W of $\mathbb{R}^n$. Define $\mathbf{v}_1 = \mathbf{x}_1$ and, for $k = 2, \ldots, p$,

$$\mathbf{v}_k = \mathbf{x}_k - \frac{\mathbf{x}_k \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \cdots - \frac{\mathbf{x}_k \cdot \mathbf{v}_{k-1}}{\mathbf{v}_{k-1} \cdot \mathbf{v}_{k-1}}\mathbf{v}_{k-1}.$$

Then $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is an orthogonal basis of W, and for each k,

$$\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \text{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}.$$

The idea

Let $W_k = \text{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$. If $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$ is already an orthogonal basis of W_{k-1}, the subtracted sum is $\text{proj}_{W_{k-1}} \mathbf{x}_k$ by (W1), so $\mathbf{v}_k$ is orthogonal to W_{k-1}. It is not $\mathbf{0}$, because $\mathbf{x}_k$ is not in W_{k-1}. Build the basis one vector at a time.

Proof. 1. Let $W_k = \text{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$; it has dimension k because the $\mathbf{x}_j$ are independent. We show that for each k, the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is an orthogonal basis of W_k. For $k = 1$ this holds because $\mathbf{v}_1 = \mathbf{x}_1 \neq \mathbf{0}$.

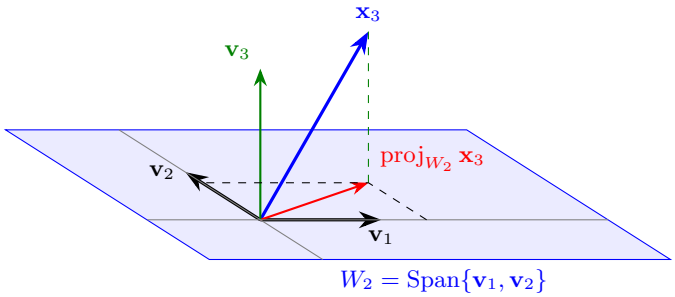
2. Suppose it holds for some $k < p$. Then by (W1) the vector subtracted from $\mathbf{x}_{k+1}$ in the definition of $\mathbf{v}_{k+1}$ is $\text{proj}_{W_k} \mathbf{x}_{k+1}$, so $\mathbf{v}_{k+1} = \mathbf{x}_{k+1} - \text{proj}_{W_k} \mathbf{x}_{k+1}$ is orthogonal to W_k, and in particular to $\mathbf{v}_1, \ldots, \mathbf{v}_k$.

3. The vector $\mathbf{v}_{k+1}$ lies in W_{k+1}: $\mathbf{x}_{k+1}$ is in W_{k+1}, $\text{proj}_{W_k} \mathbf{x}_{k+1}$ is in $W_k \subseteq W_{k+1}$, and a subspace contains differences of its vectors.

4. The vector $\mathbf{v}_{k+1}$ is not $\mathbf{0}$: otherwise $\mathbf{x}_{k+1} = \text{proj}_{W_k} \mathbf{x}_{k+1}$ would lie in $W_k = \text{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$, contradicting the independence of $\mathbf{x}_1, \ldots, \mathbf{x}_{k+1}$.

5. So $\mathbf{v}_1, \ldots, \mathbf{v}_{k+1}$ are $k+1$ nonzero orthogonal vectors in the $(k+1)$-dimensional space W_{k+1}. By (W3) they are independent and form a basis of W_{k+1}. Repeating up to $k = p$ proves the theorem; the equality of spans is the statement that $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is a basis of W_k.

□



The third step: $\mathrm{proj}_{W_2}\,\mathbf{x}_3$ is the sum of the projections of $\mathbf{x}_3$ onto the lines through $\mathbf{v}_1$ and $\mathbf{v}_2$ (gray); the dashed segments join $\mathrm{proj}_{W_2}\,\mathbf{x}_3$ to those two projections. $\mathbf{v}_3 = \mathbf{x}_3 - \mathrm{proj}_{W_2}\,\mathbf{x}_3$ is perpendicular to the plane W_2.

Example (three vectors). $\mathbf{x}_1 = (1,1,0,1)$, $\mathbf{x}_2 = (2,0,1,1)$, $\mathbf{x}_3 = (1,2,2,2)$.

1. $\mathbf{v}_1 = \mathbf{x}_1$, with $\mathbf{v}_1 \cdot \mathbf{v}_1 = 3$.
2. $\mathbf{x}_2 \cdot \mathbf{v}_1 = 2+0+0+1 = 3$, so the coefficient is $\frac{3}{3} = 1$ and $\mathbf{v}_2 = \mathbf{x}_2 - \mathbf{v}_1 = (1,-1,1,0)$, with $\mathbf{v}_2 \cdot \mathbf{v}_2 = 3$.
3. $\mathbf{x}_3 \cdot \mathbf{v}_1 = 5$ and $\mathbf{x}_3 \cdot \mathbf{v}_2 = 1$, so
$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{5}{3}\mathbf{v}_1 - \frac{1}{3}\mathbf{v}_2 = (1,2,2,2) - \left(\frac{5}{3},\frac{5}{3},0,\frac{5}{3}\right) - \left(\frac{1}{3},-\frac{1}{3},\frac{1}{3},0\right) = \left(-1,\ \frac{2}{3},\ \frac{5}{3},\ \frac{1}{3}\right).$$
4. To avoid fractions in later computations, use $3\mathbf{v}_3 = (-3,2,5,1)$, which spans the same line as $\mathbf{v}_3$. Its dot products with $\mathbf{v}_1$ and $\mathbf{v}_2$ are $-3+2+1=0$ and $-3-2+5=0$.

Orthonormal bases. Divide each $\mathbf{v}_k$ by its length at the end. In the example, $\|\mathbf{v}_1\| = \|\mathbf{v}_2\| = \sqrt{3}$ and $\|(-3,2,5,1)\| = \sqrt{39}$, so an orthonormal basis of W is
$$\frac{1}{\sqrt{3}}(1,1,0,1), \qquad \frac{1}{\sqrt{3}}(1,-1,1,0), \qquad \frac{1}{\sqrt{39}}(-3,2,5,1).$$
Normalizing only at the end avoids carrying square roots through the projections.

Every subspace has an orthogonal basis. Every nonzero subspace W of $\mathbb{R}^n$ has a basis (Week 6, Lay 4.5), and Gram–Schmidt turns it into an orthogonal basis; normalizing gives an orthonormal basis. So the orthogonal decomposition theorem and the best approximation theorem (W1, W2) apply to every subspace of $\mathbb{R}^n$, and $\mathrm{proj}_W\,\mathbf{y}$ is defined for every W. The map $\mathbf{y} \mapsto \mathrm{proj}_W\,\mathbf{y}$ is linear, because each coefficient $(\mathbf{y}\cdot\mathbf{u}_j)/(\mathbf{u}_j\cdot\mathbf{u}_j)$ in (W1) is linear in $\mathbf{y}$.

Shortcuts

- **Clear fractions as you go**: replace $\mathbf{v}_k$ by a multiple that has integer entries before using it in the next projection, and confirm with its dot products. Reason: the projection onto a line does not depend on which nonzero vector spans the line (Week 9, projection onto a line). Instance: $3\mathbf{v}_3 = (-3,2,5,1)$ above.
- **Check each new $\mathbf{v}_k$**: its dot product with every earlier $\mathbf{v}_j$ must be 0. Instance: the checks in the three-vector example.
- **Scaling keeps an orthogonal set orthogonal**: if $\mathbf{v}_1,\dots,\mathbf{v}_p$ are orthogonal and $c_1,\dots,c_p$ are nonzero scalars, then $c_1\mathbf{v}_1,\dots,c_p\mathbf{v}_p$ are orthogonal and nonzero vectors stay nonzero. Reason: for $i \neq j$,
$$(c_i\mathbf{v}_i)\cdot(c_j\mathbf{v}_j) = c_ic_j(\mathbf{v}_i\cdot\mathbf{v}_j) = 0.$$
So each vector of a Gram–Schmidt result may be rescaled separately, and the result is still an orthogonal basis of the same space (nonzero orthogonal vectors are independent, and $c_j \neq 0$ keeps the span the same). Instance: $(1,1,0)$ and $(\frac{1}{2},-\frac{1}{2},1)$ are orthogonal, and so are $(1,1,0)$ and $2(\frac{1}{2},-\frac{1}{2},1) = (1,-1,2)$: $1-1+0=0$.
- **Already orthogonal**: if $\mathbf{x}_k$ is orthogonal to the earlier $\mathbf{v}_j$, every subtracted term is 0 and $\mathbf{v}_k = \mathbf{x}_k$. Instance: $\mathbf{x}_1 = (1,0,1)$, $\mathbf{x}_2 = (1,0,-1)$ have dot product 0, so $\mathbf{v}_2 = \mathbf{x}_2$.

Traps

- Project onto the new vectors $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$, not onto the original $\mathbf{x}_j$. For $\mathbf{x}_1 = (1,0,0)$, $\mathbf{x}_2 = (1,1,0)$, $\mathbf{x}_3 = (1,1,1)$, Gram–Schmidt gives $(1,0,0)$, $(0,1,0)$, $(0,0,1)$. Subtracting from $\mathbf{x}_3$ its projections onto $\mathbf{x}_1$ and $\mathbf{x}_2$ instead gives
$$(1,1,1) - 1 \cdot (1,0,0) - \frac{2}{2}(1,1,0) = (-1,0,1),$$
which is not orthogonal to $\mathbf{x}_1$.
- If Gram–Schmidt produces $\mathbf{v}_k = \mathbf{0}$, the starting vectors were dependent: $\mathbf{x}_k$ is in the span of the earlier ones. Instance: $\mathbf{x}_1 = (1,2)$, $\mathbf{x}_2 = (2,4)$ give $\mathbf{v}_2 = (2,4) - 2(1,2) = \mathbf{0}$.

§2.2 The QR factorization (Lay 6.4)

The question. Gram–Schmidt applied to the columns $(1,1,0,1)$, $(2,0,1,1)$ of a 4×2 matrix A writes each column as a combination of the new orthonormal vectors. Can these combinations be written as a matrix factorization $A = QR$, and what is R?

Theorem (QR factorization (Lay Thm. 12))
If A is an $m \times n$ matrix with linearly independent columns, then $A = QR$, where Q is $m \times n$ with orthonormal columns forming a basis of $\operatorname{Col} A$, and R is $n \times n$, upper triangular and invertible, with positive diagonal entries.

The idea

Gram–Schmidt gives $\operatorname{Span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} = \operatorname{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, so column k of A is a combination of the first k columns of Q only. Writing these combinations as the columns of R makes R upper triangular.

Proof. 1. Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be the columns of A; they are a basis of $\operatorname{Col} A$. Apply Gram–Schmidt to get $\mathbf{v}_1, \dots, \mathbf{v}_n$, and let $\mathbf{q}_k = \mathbf{v}_k / \|\mathbf{v}_k\|$ and $Q = [\,\mathbf{q}_1 \cdots \mathbf{q}_n\,]$. Then Q has orthonormal columns that form a basis of $\operatorname{Col} A$.

2. Solving the formula of the Gram–Schmidt process (Thm. 11) for $\mathbf{x}_k$ gives $\mathbf{x}_k = \mathbf{v}_k + (\text{a combination of } \mathbf{v}_1, \dots, \mathbf{v}_{k-1})$. Since $\mathbf{v}_j = \|\mathbf{v}_j\|\,\mathbf{q}_j$, this says
$$\mathbf{x}_k = r_{1k}\mathbf{q}_1 + \cdots + r_{kk}\mathbf{q}_k \qquad \text{with } r_{kk} = \|\mathbf{v}_k\| > 0.$$

3. Let column k of R be $(r_{1k}, \dots, r_{kk}, 0, \dots, 0)$. Then Q times column k of R is $\mathbf{x}_k$, so $QR = A$. The matrix R is upper triangular with positive diagonal entries, so it is invertible (W6).

□

Computing R. Multiply $A = QR$ on the left by Q^{T} and use $Q^{\mathsf{T}}Q = I$ (W4): $R = Q^{\mathsf{T}}A$. Its entries are $r_{ik} = \mathbf{q}_i \cdot \mathbf{x}_k$.

Example. Let A have columns $\mathbf{x}_1 = (1,1,0,1)$ and $\mathbf{x}_2 = (2,0,1,1)$. From the example above, $\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1,1,0,1)$ and $\mathbf{q}_2 = \frac{1}{\sqrt{3}}(1,-1,1,0)$. Then
$$r_{11} = \mathbf{q}_1 \cdot \mathbf{x}_1 = \frac{1+1+0+1}{\sqrt{3}} = \sqrt{3}, \qquad r_{12} = \mathbf{q}_1 \cdot \mathbf{x}_2 = \frac{2+0+0+1}{\sqrt{3}} = \sqrt{3},$$
$$r_{22} = \mathbf{q}_2 \cdot \mathbf{x}_2 = \frac{2-0+1+0}{\sqrt{3}} = \sqrt{3},$$
so
$$Q = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{3} \end{bmatrix}.$$
Check column 2:
$$\sqrt{3}\,\mathbf{q}_1 + \sqrt{3}\,\mathbf{q}_2 = (1,1,0,1) + (1,-1,1,0) = (2,0,1,1) = \mathbf{x}_2.$$

Shortcuts

- $R = Q^{\mathsf{T}}A$: once Q is known, R is a matrix product; no system to solve. Reason: $Q^{\mathsf{T}}Q = I$. Instance: the example above.
- **Diagonal of R**: $r_{kk} = \|\mathbf{v}_k\|$, the lengths found while normalizing. Reason: step 2 of the proof. Instance: $r_{11} = \|\mathbf{v}_1\| = \sqrt{3}$.
- **Same column space**: if $A = QR$ with R invertible, then $\operatorname{Col} A = \operatorname{Col} Q$. Reason: $A\mathbf{x} = Q(R\mathbf{x})$, and $Q\mathbf{z} = A(R^{-1}\mathbf{z})$.
- **Check**: QR must reproduce a column of A. Instance: the check of column 2.

Traps

- The entries of R use the unit vectors: $r_{ik} = \mathbf{q}_i \cdot \mathbf{x}_k$, not $\mathbf{v}_i \cdot \mathbf{x}_k$. In the example, $\mathbf{v}_1 \cdot \mathbf{x}_1 = 3$, but $r_{11} = \mathbf{q}_1 \cdot \mathbf{x}_1 = \sqrt{3}$.
- QR as stated needs independent columns. For the columns $(1,1)$, $(2,2)$, Gram–Schmidt gives $\mathbf{v}_2 = \mathbf{0}$, which cannot be normalized.

§2.3 Least-squares problems (Lay 6.5)

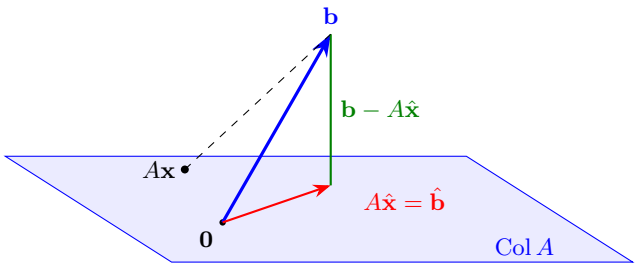
The question. No line $y = c_0 + c_1 x$ passes through all three points $(0,1)$, $(1,2)$, $(2,4)$: the system $c_0 + c_1 x_i = y_i$ has no solution. When $A\mathbf{x} = \mathbf{b}$ has no solution, what is the best substitute, and how do we compute it?

Definition. For an $m \times n$ matrix A and $\mathbf{b} \in \mathbb{R}^m$, a **least-squares solution** of $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ with

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \le \|\mathbf{b} - A\mathbf{x}\| \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

The number $\|\mathbf{b} - A\hat{\mathbf{x}}\|$ is the **least-squares error**.

- The vectors $A\mathbf{x}$ are exactly the vectors of $\operatorname{Col} A$. So we want the point of $\operatorname{Col} A$ closest to $\mathbf{b}$, which is $\hat{\mathbf{b}} = \operatorname{proj}_{\operatorname{Col} A} \mathbf{b}$ (W2), and no other point of $\operatorname{Col} A$ is as close. So the least-squares solutions are exactly the $\hat{\mathbf{x}}$ with $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$; this equation is consistent because $\hat{\mathbf{b}} \in \operatorname{Col} A$.
- The residual $\mathbf{b} - A\hat{\mathbf{x}} = \mathbf{b} - \hat{\mathbf{b}}$ is orthogonal to $\operatorname{Col} A$ (W1), so by (W4) $A^{\mathsf{T}}(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$. That is the next theorem.



$A\hat{\mathbf{x}}$ is the point of $\operatorname{Col} A$ closest to $\mathbf{b}$, and the residual $\mathbf{b} - A\hat{\mathbf{x}}$ is perpendicular to $\operatorname{Col} A$. For any other $A\mathbf{x}$ in $\operatorname{Col} A$, the dashed distance $\|\mathbf{b} - A\mathbf{x}\|$ is larger.

Theorem (Normal equations (Lay Thm. 13))

The least-squares solutions of $A\mathbf{x} = \mathbf{b}$ are exactly the solutions of the **normal equations**

$$A^\mathsf{T}A\mathbf{x} = A^\mathsf{T}\mathbf{b}.$$

In particular, this system is always consistent.

Proof. 1. **Assume $\hat{\mathbf{x}}$ is a least-squares solution.** Then $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$: otherwise $A\hat{\mathbf{x}}$ would be a point of $\operatorname{Col} A$ other than $\hat{\mathbf{b}}$, and (W2) would give $\|\mathbf{b} - \hat{\mathbf{b}}\| < \|\mathbf{b} - A\hat{\mathbf{x}}\|$, so $\hat{\mathbf{x}}$ would not minimize. So $\mathbf{b} - A\hat{\mathbf{x}} = \mathbf{b} - \hat{\mathbf{b}}$ is in $(\operatorname{Col} A)^\perp = \operatorname{Nul} A^\mathsf{T}$ (W1, W4). Hence $A^\mathsf{T}\mathbf{b} - A^\mathsf{T}A\hat{\mathbf{x}} = \mathbf{0}$.

2. **Conversely, assume $A^\mathsf{T}A\hat{\mathbf{x}} = A^\mathsf{T}\mathbf{b}$.** Then $\mathbf{b} - A\hat{\mathbf{x}}$ is in $\operatorname{Nul} A^\mathsf{T} = (\operatorname{Col} A)^\perp$ (W4), so
$$\mathbf{b} = A\hat{\mathbf{x}} + (\mathbf{b} - A\hat{\mathbf{x}})$$
writes $\mathbf{b}$ as a vector in $\operatorname{Col} A$ plus a vector in $(\operatorname{Col} A)^\perp$.

3. By uniqueness of that decomposition (W1), $A\hat{\mathbf{x}} = \operatorname{proj}_{\operatorname{Col} A}\mathbf{b}$, so $\hat{\mathbf{x}}$ is a least-squares solution.

4. The equation $A\mathbf{x} = \hat{\mathbf{b}}$ is consistent because $\hat{\mathbf{b}} \in \operatorname{Col} A$, so least-squares solutions exist, and by step 1 they solve the normal equations.

□

Example (fitting a line). Find the line $y = c_0 + c_1 x$ that best fits the points $(0,1)$, $(1,2)$, $(2,4)$. Requiring the line to pass through all three points gives three equations $c_0 + c_1 x_i = y_i$ in two unknowns, which have no solution:

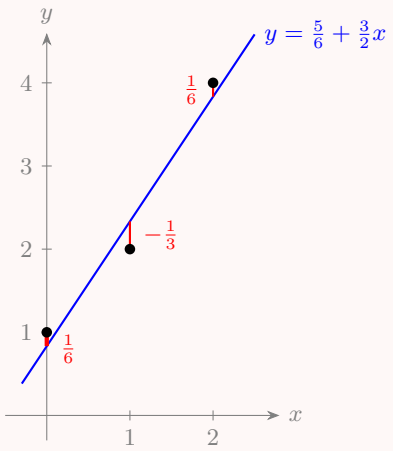
$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \qquad A^\mathsf{T}A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad A^\mathsf{T}\mathbf{b} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}.$$

The determinant of $A^\mathsf{T}A$ is 6, so

$$\hat{\mathbf{x}} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} 5/6 \\ 3/2 \end{bmatrix}, \qquad \mathbf{b} - A\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 5/6 \\ 7/3 \\ 23/6 \end{bmatrix} = \begin{bmatrix} 1/6 \\ -1/3 \\ 1/6 \end{bmatrix}.$$

The best line is $y = \frac{5}{6} + \frac{3}{2}x$. Check: $A^\mathsf{T}(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$. The least-squares error is

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| = \frac{1}{6}\sqrt{1+4+1} = \frac{\sqrt{6}}{6}.$$



The least-squares line minimizes the sum of the squares of the vertical gaps (red), which are the entries $\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}$ of $\mathbf{b} - A\hat{\mathbf{x}}$.

Theorem (When the least-squares solution is unique (Lay Thm. 14))

For an $m \times n$ matrix A, the following are equivalent: (a) $A\mathbf{x} = \mathbf{b}$ has exactly one least-squares solution for each $\mathbf{b} \in \mathbb{R}^m$; (b) the columns of A are linearly independent; (c) $A^\mathsf{T}A$ is invertible. In that case

$$\hat{\mathbf{x}} = (A^\mathsf{T}A)^{-1}A^\mathsf{T}\mathbf{b}.$$

Proof. 1. (b)⇔(c). $A^\mathsf{T}A$ is $n \times n$, and $\operatorname{Nul} A^\mathsf{T}A = \operatorname{Nul} A$ (W5). So $A^\mathsf{T}A$ is invertible if and only if $\operatorname{Nul} A^\mathsf{T}A = \{\mathbf{0}\}$, if and only if $\operatorname{Nul} A = \{\mathbf{0}\}$, which says the columns of A are independent.

2. (c)⇒(a). If $A^\mathsf{T}A$ is invertible, the normal equations have the single solution $(A^\mathsf{T}A)^{-1}A^\mathsf{T}\mathbf{b}$.

3. (a)⇒(b), by contrapositive. **Assume the columns are dependent**: $A\mathbf{w} = \mathbf{0}$ for some $\mathbf{w} \neq \mathbf{0}$. If $\hat{\mathbf{x}}$ is a least-squares solution, then $A(\hat{\mathbf{x}} + \mathbf{w}) = A\hat{\mathbf{x}}$, so $\hat{\mathbf{x}} + \mathbf{w}$ is a second least-squares solution, and (a) fails.

□

Two consequences: $\operatorname{rank} A^\mathsf{T}A = \operatorname{rank} A$, because both matrices have n columns and the same null space (W5, W9). And independent columns require $m \geq n$, since n independent vectors in $\mathbb{R}^m$ need $n \leq m$.

- **Dependent columns.** The projection $\hat{\mathbf{b}}$ is always unique, but $\hat{\mathbf{x}}$ is not when $\operatorname{Nul} A \neq \{\mathbf{0}\}$. Example: A and $\mathbf{b}$ below, with the normal equations:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \qquad \left[\begin{array}{cc|c} 2 & 4 & 4 \\ 4 & 8 & 8 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 2 & 4 & 4 \\ 0 & 0 & 0 \end{array}\right].$$

  So the least-squares solutions are all $\hat{\mathbf{x}}$ with $\hat{x}_1 = 2 - 2\hat{x}_2$; in parametric vector form,

$$\hat{\mathbf{x}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \hat{x}_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

  Each gives $A\hat{\mathbf{x}} = (2, 2, 0)$.

- **Orthogonal columns.** If the columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ of A are orthogonal and nonzero, then (W1) gives $\hat{\mathbf{b}}$ directly, and its weights are the entries of $\hat{\mathbf{x}}$:

$$\hat{x}_j = \frac{\mathbf{b} \cdot \mathbf{a}_j}{\mathbf{a}_j \cdot \mathbf{a}_j}.$$

  Example: $\mathbf{a}_1 = (1, 1, 1, 1)$, $\mathbf{a}_2 = (-3, -1, 1, 3)$, $\mathbf{b} = (0, 1, 1, 3)$. Then, with no system to solve,

$$\hat{x}_1 = \frac{0 + 1 + 1 + 3}{4} = \frac{5}{4}, \qquad \hat{x}_2 = \frac{0 - 1 + 1 + 9}{20} = \frac{9}{20}.$$

  (For a line fit, the column of x-values is orthogonal to the column of ones exactly when the x-values have mean 0, as here.)

Theorem (Least squares with QR (Lay Thm. 15))

If A has independent columns and $A = QR$ as in Thm. 12, then for each $\mathbf{b}$ the unique least-squares solution of $A\mathbf{x} = \mathbf{b}$ is the solution of

$$R\hat{\mathbf{x}} = Q^\mathsf{T}\mathbf{b}, \qquad \text{that is,} \qquad \hat{\mathbf{x}} = R^{-1}Q^\mathsf{T}\mathbf{b}.$$

Proof. 1. Since $(QR)^\mathsf{T} = R^\mathsf{T}Q^\mathsf{T}$ and $Q^\mathsf{T}Q = I$ (W4, W6),

$$A^\mathsf{T}A = R^\mathsf{T}Q^\mathsf{T}QR = R^\mathsf{T}R, \qquad A^\mathsf{T}\mathbf{b} = R^\mathsf{T}Q^\mathsf{T}\mathbf{b}.$$

2. So the normal equations are $R^\mathsf{T}R\mathbf{x} = R^\mathsf{T}Q^\mathsf{T}\mathbf{b}$.

3. R^T is invertible because R is (W6). Multiplying by $(R^\mathsf{T})^{-1}$ gives the equivalent system $R\mathbf{x} = Q^\mathsf{T}\mathbf{b}$, whose unique solution is $R^{-1}Q^\mathsf{T}\mathbf{b}$.

□

Because R is upper triangular, solve $R\hat{\mathbf{x}} = Q^{\mathsf{T}}\mathbf{b}$ by back substitution instead of inverting R. Example: with Q, R from the QR example and $\mathbf{b} = (3, 1, 0, 2)$,

$$Q^{\mathsf{T}}\mathbf{b} = \frac{1}{\sqrt{3}}\begin{bmatrix}6\\2\end{bmatrix}, \qquad \begin{bmatrix}\sqrt{3} & \sqrt{3}\\ 0 & \sqrt{3}\end{bmatrix}\hat{\mathbf{x}} = \frac{1}{\sqrt{3}}\begin{bmatrix}6\\2\end{bmatrix}.$$

The second row gives $\sqrt{3}\,\hat{x}_2 = 2/\sqrt{3}$, so $\hat{x}_2 = 2/3$. The first row, $\sqrt{3}\,\hat{x}_1 + \sqrt{3}\,\hat{x}_2 = 6/\sqrt{3}$, divided by $\sqrt{3}$, is $\hat{x}_1 + \hat{x}_2 = 2$, so $\hat{x}_1 = 4/3$. The normal equations give the same answer; subtracting the first, $3\hat{x}_1 + 3\hat{x}_2 = 6$, from the second, $3\hat{x}_1 + 6\hat{x}_2 = 8$, gives $3\hat{x}_2 = 2$:

$$A^{\mathsf{T}}A = \begin{bmatrix}3 & 3\\ 3 & 6\end{bmatrix}, \qquad A^{\mathsf{T}}\mathbf{b} = \begin{bmatrix}6\\8\end{bmatrix}, \qquad \hat{x}_2 = \frac{8-6}{3} = \frac{2}{3}, \quad \hat{x}_1 = 2 - \hat{x}_2 = \frac{4}{3}.$$

Shortcuts

- 2×2 **normal equations**: entry (i, j) of $A^{\mathsf{T}}A$ is $\mathbf{a}_i \cdot \mathbf{a}_j$, so only three dot products are needed (the matrix is symmetric); then use the inverse formula (W8). Condition: two independent columns. Instance: the line fit above.
- **Check**: $A^{\mathsf{T}}(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$, the residual is orthogonal to every column. Instance: the line fit.
- **Orthogonal columns**: $\hat{x}_j = (\mathbf{b} \cdot \mathbf{a}_j)/(\mathbf{a}_j \cdot \mathbf{a}_j)$, no system. Condition: nonzero orthogonal columns. Instance: the example with $\mathbf{a}_1 = (1, 1, 1, 1)$, $\mathbf{a}_2 = (-3, -1, 1, 3)$.
- **If $\mathbf{b} \in \operatorname{Col} A$**: $\hat{\mathbf{b}} = \mathbf{b}$, so the least-squares solutions are the ordinary solutions and the error is 0. Reason: (W1), uniqueness of the decomposition $\mathbf{b} = \mathbf{b} + \mathbf{0}$. Instance: for the line-fit A and $\mathbf{b} = (1, 2, 3)$ (the points lie on $y = 1 + x$), $\hat{\mathbf{x}} = (1, 1)$ and the error is 0.
- **With a QR factorization**: solve $R\hat{\mathbf{x}} = Q^{\mathsf{T}}\mathbf{b}$ from the bottom row up. Instance: the example above.

Traps

- The normal equations are $A^{\mathsf{T}}A\mathbf{x} = A^{\mathsf{T}}\mathbf{b}$. Cancelling A^{T} is not allowed: it is usually not square, and cancelling would give back the inconsistent system $A\mathbf{x} = \mathbf{b}$ (in the line fit, three equations with no solution).
- The formula $\hat{\mathbf{x}} = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\mathbf{b}$ needs independent columns (Thm. 14). In the dependent-columns example, $A^{\mathsf{T}}A$ has determinant $2 \cdot 8 - 4 \cdot 4 = 0$, so $(A^{\mathsf{T}}A)^{-1}$ does not exist, yet least-squares solutions exist (a whole line of them). Also, $(A^{\mathsf{T}}A)^{-1}$ is not $A^{-1}(A^{\mathsf{T}})^{-1}$ unless A is square and invertible; when it is, $\hat{\mathbf{x}} = A^{-1}\mathbf{b}$, the ordinary solution.
- $\hat{\mathbf{x}}$, $A\hat{\mathbf{x}}$ and the error are different things. In the line fit, $\hat{\mathbf{x}} = (\frac{5}{6}, \frac{3}{2})$ is in $\mathbb{R}^2$, $A\hat{\mathbf{x}} = \hat{\mathbf{b}} = (\frac{5}{6}, \frac{7}{3}, \frac{23}{6})$ is in $\mathbb{R}^3$, and the error is $\|\mathbf{b} - A\hat{\mathbf{x}}\| = \frac{\sqrt{6}}{6}$, not $\|\hat{\mathbf{x}}\|$.

§2.4 Inner product spaces (Lay 6.7)

The question. Which linear function $c_0 + c_1 t$ is closest to t^2 on the interval $[0, 1]$? To answer, the space of functions needs lengths and projections. The proofs of the Pythagorean theorem, the orthogonal decomposition theorem (Week 9, Thm. 8), best approximation and Gram–Schmidt used only a few algebraic rules of the dot product. What are the rules, so that we can use them on other vector spaces?

Definition. An **inner product** on a vector space V assigns to each pair $\mathbf{u}, \mathbf{v} \in V$ a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and scalars c:

(1) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$;

(2) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$;

(3) $\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$;

(4) $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ only for $\mathbf{u} = \mathbf{0}$.

A vector space with an inner product is an **inner product space**. Length, distance and orthogonality are defined as in $\mathbb{R}^n$:

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}, \qquad \operatorname{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|, \qquad \mathbf{u} \perp \mathbf{v} \iff \langle \mathbf{u}, \mathbf{v}\rangle = 0.$$

Rules that follow from the axioms.

- **The second slot is linear too**: by (1), then (2) or (3), then (1) again,

$$\langle \mathbf{u}, \mathbf{v} + \mathbf{w}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle, \qquad \langle \mathbf{u}, c\mathbf{v}\rangle = c\langle \mathbf{u}, \mathbf{v}\rangle.$$

- **Scaling rule**: $\langle c\mathbf{u}, c\mathbf{u}\rangle = c^2\langle \mathbf{u}, \mathbf{u}\rangle$ by (3) in each slot, so $\|c\mathbf{u}\| = |c|\,\|\mathbf{u}\|$.
- **Expansion rule**: by linearity in both slots and (1),

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v}\rangle + \|\mathbf{v}\|^2, \qquad \|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2\langle \mathbf{u}, \mathbf{v}\rangle + \|\mathbf{v}\|^2.$$

  Adding and subtracting these gives the parallelogram law and the polarization identity:

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2, \qquad \langle \mathbf{u}, \mathbf{v}\rangle = \frac{1}{4}\big(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2\big).$$

  The Pythagorean theorem (W7) follows from the first expansion exactly as in $\mathbb{R}^n$.

Examples.

- **A weighted inner product on $\mathbb{R}^2$:** $\langle \mathbf{u}, \mathbf{v}\rangle = 2u_1v_1 + 3u_2v_2$. Axioms (1)–(3) are direct. For (4), $2u_1^2 + 3u_2^2 \geq 0$, with equality only if $u_1 = u_2 = 0$. In this inner product $(1, 1)$ and $(3, -2)$ are orthogonal ($6 - 6 = 0$), although their dot product is 1. Orthogonality depends on the inner product.
- **Evaluation on $\mathbb{P}_n$:** pick $n + 1$ distinct numbers $t_0, \ldots, t_n$ and set

$$\langle p, q\rangle = p(t_0)q(t_0) + \cdots + p(t_n)q(t_n).$$

  This is the dot product of the value vectors $(p(t_0), \ldots, p(t_n))$ and $(q(t_0), \ldots, q(t_n))$, so (1)–(3) hold. For (4): if $\langle p, p\rangle = 0$, then $p(t_i) = 0$ for $n + 1$ distinct t_i; a nonzero polynomial of degree at most n has at most n roots (a fact from algebra), so $p = 0$.
- **The integral inner product on $C[a, b]$** (continuous functions on $[a, b]$):

$$\langle f, g\rangle = \int_a^b f(t)g(t)\,dt.$$

  Axioms (1)–(3) are properties of integrals. For (4): $f(t)^2$ is continuous and nonnegative, and a continuous nonnegative function g whose integral over $[a, b]$ is 0 is identically 0 (calculus): if $g(t_0) > 0$, then by continuity $g > g(t_0)/2$ on some interval around t_0, and the integral would be positive.

Gram–Schmidt and projections in an inner product space. The proofs of (W1), (W2), (W3), (W7) and Thm. 11 use only axioms (1)–(4) and the rules above, so they hold with $\langle \cdot, \cdot\rangle$ in place of the dot product, for any finite-dimensional subspace W. Example: on $\mathbb{P}_2$ with evaluation at $-1, 0, 1$, apply Gram–Schmidt to $1, t, t^2$.

1. The value vectors are $1 \mapsto (1, 1, 1)$, $t \mapsto (-1, 0, 1)$, $t^2 \mapsto (1, 0, 1)$.
2. $\langle t, 1\rangle = -1 + 0 + 1 = 0$, so t is already orthogonal to 1: $p_0 = 1$, $p_1 = t$.
3. $\langle t^2, 1\rangle = 2$, $\langle 1, 1\rangle = 3$, $\langle t^2, t\rangle = 0$, so $p_2 = t^2 - \frac{2}{3}$, with value vector $(\frac{1}{3}, -\frac{2}{3}, \frac{1}{3})$, orthogonal to $(1, 1, 1)$ and $(-1, 0, 1)$.
4. The same numbers give the best approximation of t^2 by a linear function in this inner product: $\operatorname{proj} t^2 = \frac{2}{3} \cdot 1 + 0 \cdot t = \frac{2}{3}$, the constant that matches the values $1, 0, 1$ best.

Best approximation by a polynomial. In $C[0, 1]$ with the integral inner product, find the linear function closest to t^2.

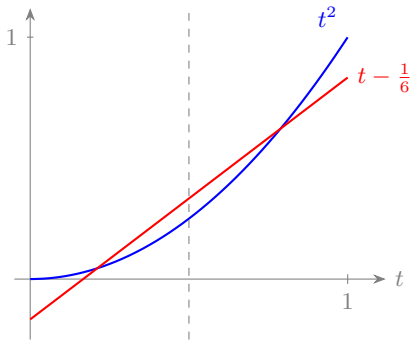
1. An orthogonal basis of $\mathbb{P}_1$: $\langle t, 1\rangle = \int_0^1 t\,dt = \frac{1}{2}$ and $\langle 1, 1\rangle = 1$, so Gram–Schmidt gives 1 and $t - \frac{1}{2}$.
2. The inner products needed are

$$\langle t^2, 1\rangle = \frac{1}{3}, \qquad \langle t^2, t - \tfrac{1}{2}\rangle = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}, \qquad \langle t - \tfrac{1}{2}, t - \tfrac{1}{2}\rangle = \int_0^1 \big(t - \tfrac{1}{2}\big)^2 dt = \frac{1}{12}.$$

3. So
$$\operatorname{proj}_{\mathbb{P}_1} t^2 = \frac{1}{3}\cdot 1 + \frac{1/12}{1/12}\left(t - \frac{1}{2}\right) = t - \frac{1}{6}.$$

4. Its distance from t^2, by the Pythagorean theorem for $t^2 = \hat{p} + (t^2 - \hat{p})$ with $\hat{p} = t - \frac{1}{6}$ and $\|\hat{p}\|^2 = \frac{1}{9}\langle 1,1\rangle + \langle t-\frac{1}{2}, t-\frac{1}{2}\rangle$:
$$\|t^2 - \hat{p}\|^2 = \|t^2\|^2 - \|\hat{p}\|^2 = \int_0^1 t^4\,dt - \left(\frac{1}{9} + \frac{1}{12}\right) = \frac{1}{5} - \frac{7}{36} = \frac{1}{180}.$$

So the distance is $1/\sqrt{180}$, smaller than for any other linear function (W2).



t^2 (blue) and its best linear approximation $t - \frac{1}{6}$ in the integral inner product on $[0,1]$ (red). The gap $g = t^2 - t + \frac{1}{6}$ is orthogonal to 1 and to $t - \frac{1}{2}$: $\int_0^1 g\,dt = 0$ and $\int_0^1 g\,(t-\frac{1}{2})\,dt = 0$. The first condition forces g to change sign, so the line crosses the parabola twice. The dashed line marks $t = \frac{1}{2}$.

Theorem (Cauchy–Schwarz inequality (Lay Thm. 16))
In an inner product space, $|\langle \mathbf{u}, \mathbf{v}\rangle| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$ for all $\mathbf{u}, \mathbf{v}$.

The idea

Project $\mathbf{v}$ onto the line through $\mathbf{u}$. The projection has length $|\langle \mathbf{v}, \mathbf{u}\rangle|/\|\mathbf{u}\|$, and by Pythagoras a leg of a right triangle is no longer than the hypotenuse $\mathbf{v}$.

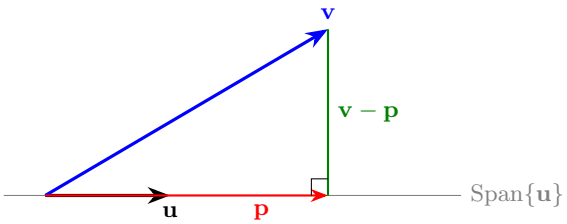
Proof. 1. If $\mathbf{u} = \mathbf{0}$, both sides are 0. Otherwise let $W = \operatorname{Span}\{\mathbf{u}\}$ and
$$\mathbf{p} = \operatorname{proj}_W \mathbf{v} = \frac{\langle \mathbf{v}, \mathbf{u}\rangle}{\langle \mathbf{u}, \mathbf{u}\rangle}\,\mathbf{u}.$$

2. The vector $\mathbf{v} - \mathbf{p}$ is orthogonal to $\mathbf{p}$ (W1), so the Pythagorean theorem gives $\|\mathbf{v}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{v} - \mathbf{p}\|^2 \ge \|\mathbf{p}\|^2$.

3. By the scaling rule, $\|c\mathbf{u}\| = |c|\,\|\mathbf{u}\|$, so
$$\|\mathbf{p}\| = \frac{|\langle \mathbf{v}, \mathbf{u}\rangle|}{\|\mathbf{u}\|^2}\,\|\mathbf{u}\| = \frac{|\langle \mathbf{u}, \mathbf{v}\rangle|}{\|\mathbf{u}\|}.$$

4. Combining, $|\langle \mathbf{u}, \mathbf{v}\rangle|/\|\mathbf{u}\| \le \|\mathbf{v}\|$. Multiply by $\|\mathbf{u}\|$.

□



Cauchy–Schwarz: $\|\mathbf{p}\| = |\langle \mathbf{u}, \mathbf{v}\rangle|/\|\mathbf{u}\|$ is the length of a leg of a right triangle whose hypotenuse is $\mathbf{v}$, so it is at most $\|\mathbf{v}\|$.

Theorem (Triangle inequality (Lay Thm. 17))
$\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.

Proof. 1. By the expansion rule, $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v}\rangle + \|\mathbf{v}\|^2$.

2. By Cauchy–Schwarz, $\langle \mathbf{u}, \mathbf{v}\rangle \le |\langle \mathbf{u}, \mathbf{v}\rangle| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$, so

$$\|\mathbf{u}+\mathbf{v}\|^2 \le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\,\|\mathbf{v}\| + \|\mathbf{v}\|^2 = (\|\mathbf{u}\| + \|\mathbf{v}\|)^2.$$

3. Both sides are squares of nonnegative numbers, so taking square roots preserves the inequality.

□

Shortcuts

- **Evaluation inner product**: compute with the value vectors; it is a dot product in $\mathbb{R}^{n+1}$. Instance: the Gram–Schmidt example on $\mathbb{P}_2$.
- **Symmetric points or intervals**: if the evaluation points are symmetric about 0 (or the interval is $[-a, a]$), an even polynomial and an odd polynomial are orthogonal. Reason: their product r is odd, so the values $r(t)$ and $r(-t) = -r(t)$ cancel in pairs, and $r(0) = -r(0) = 0$ (and $\int_{-a}^{a} r = 0$). Instance: at $-1, 0, 1$, $\langle t, 1\rangle = 0$ and $\langle t^2, t\rangle = 0$.
- **Distance by the Pythagorean theorem**: $\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = \|\mathbf{y}\|^2 - \|\hat{\mathbf{y}}\|^2$, and $\|\hat{\mathbf{y}}\|^2$ is a sum of squared coefficients times $\langle \mathbf{u}_j, \mathbf{u}_j\rangle$. Instance: $\frac{1}{5} - \frac{7}{36} = \frac{1}{180}$ for t^2 above, with no integral of a square.
- **Best approximation in a function space**: find an orthogonal basis of the subspace by Gram–Schmidt, then use the formula of the orthogonal decomposition theorem (Week 9, Thm. 8) with integrals. Instance: $t - \frac{1}{6}$ above.

Traps

- Evaluation at fewer than $n+1$ points is not an inner product on $\mathbb{P}_n$: on $\mathbb{P}_2$ with the points 0 and 1, the polynomial $t(t-1) \neq 0$ has $\langle t(t-1), t(t-1)\rangle = 0$, so axiom (4) fails.
- A weighted formula is an inner product only with positive weights: $2u_1v_1 - 3u_2v_2$ gives $\langle (0,1), (0,1)\rangle = -3 < 0$, so axiom (4) fails.
- With a weighted or integral inner product, compute $\langle \cdot, \cdot \rangle$ with its own formula; orthogonal no longer means dot product 0. Instance: $(1,1)$ and $(3,-2)$ under $2u_1v_1 + 3u_2v_2$.
- In the formula of the orthogonal decomposition theorem, divide by $\langle \mathbf{u}_j, \mathbf{u}_j\rangle$ computed in the same inner product. In the $C[0,1]$ example, $\langle t - \frac{1}{2}, t - \frac{1}{2}\rangle = \frac{1}{12}$, not the $\frac{1}{4} + 1$ that the coefficients of $t - \frac{1}{2}$ would give.

§3 Homework this week

HW9, Sections 6.4–6.5, and HW10, Section 6.7.

Section 6.4.

- **#3** (Gram–Schmidt, two vectors). $\mathbf{v}_1 = \mathbf{x}_1$, $\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1$; check $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$.
- **#7** (orthonormal basis for #3). Clear fractions in $\mathbf{v}_2$, then divide each vector by its length.
- **#11** (orthogonal basis of $\operatorname{Col} A$, 5×3). Thm. 11: project $\mathbf{x}_3$ onto $\mathbf{v}_1$ and $\mathbf{v}_2$, not onto $\mathbf{x}_1, \mathbf{x}_2$; check each new vector.
- **#15** (QR of the matrix in #11). Q = the normalized vectors from #11, then $R = Q^{\mathsf{T}}A$ (Thm. 12); check one column of QR.
- **#17** (true/false: $\{\mathbf{v}_1, \mathbf{v}_2, c\mathbf{v}_3\}$). Scaling keeps orthogonality; consider $c = 0$.
- **#20** (true/false: $\mathbf{x} \notin W \Rightarrow \mathbf{x} - \operatorname{proj}_W \mathbf{x} \neq \mathbf{0}$). If it were $\mathbf{0}$, then $\mathbf{x} = \operatorname{proj}_W \mathbf{x} \in W$.

- **#22** (true/false: columns of Q). Thm. 12 and the Same column space shortcut.

Section 6.5.

- **#3** (normal equations, 4×2). Thm. 13: $A^{\mathsf{T}}A$ is symmetric (three dot products), then the 2×2 inverse formula.
- **#5** (all least-squares solutions). Column 1 is the sum of columns 2 and 3, so row reduce $[\, A^{\mathsf{T}}A \mid A^{\mathsf{T}}\mathbf{b} \,]$ and write the solutions in parametric vector form (the Dependent columns example).
- **#11** ($\hat{\mathbf{b}}$ and $\hat{\mathbf{x}}$, 4×3). The columns are mutually orthogonal: Orthogonal columns shortcut, then $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$.
- **#13** (could $\mathbf{u}$ be a least-squares solution?). Compare $\|\mathbf{b} - A\mathbf{u}\|$ with $\|\mathbf{b} - A\mathbf{v}\|$; if $A\mathbf{v}$ is closer, $\mathbf{u}$ is not one.
- **#17, #19** (true/false: what least squares means). $\hat{\mathbf{x}}$ minimizes $\|\mathbf{b} - A\mathbf{x}\|$; by the best approximation theorem this means $A\hat{\mathbf{x}} = \operatorname{proj}_{\operatorname{Col} A} \mathbf{b}$.
- **#22** (true/false: independent columns, one solution). Thm. 14.

Section 6.7.

- **#1** ($\langle \mathbf{u}, \mathbf{v} \rangle = 4u_1v_1 + 5u_2v_2$ on $\mathbb{R}^2$). Use this formula, not the dot product; the vectors orthogonal to $\mathbf{y}$ solve the one equation $\langle \mathbf{z}, \mathbf{y} \rangle = 0$.
- **#3, #5, #7** (evaluation inner product on $\mathbb{P}_2$ at $-1, 0, 1$). Evaluation inner product shortcut: dot products of value vectors; $\|p\|$ is the length of the value vector; then $\frac{\langle q,p \rangle}{\langle p,p \rangle} p$.
- **#9** (evaluation at $-3, -1, 1, 3$; project t^2 onto $\operatorname{Span}\{1, t\}$). Symmetric points shortcut: $\langle t^2, t \rangle = 0$ and $\langle t, 1 \rangle = 0$, so only the constant term of the projection survives; $q = t^2 -$ that projection.
- **#13** ($\langle \mathbf{u}, \mathbf{v} \rangle = (A\mathbf{u}) \cdot (A\mathbf{v})$). Axioms (1)–(3) follow from the dot product and linearity of A; (4) needs $A\mathbf{u} = \mathbf{0} \Rightarrow \mathbf{u} = \mathbf{0}$, which is invertibility.
- **#16, #18** ($\|\mathbf{u} - \mathbf{v}\| = \sqrt{2}$; parallelogram law). Expansion: $\|\mathbf{u} \pm \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \pm 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$; for #18 add the two.
- **#20, #23** (true/false). Two nonzero orthogonal vectors in $\mathbb{R}^2$; axiom (4) gives $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$.
- **#26** (compare the squared mean of a, b with the mean of their squares). Thm. 16 with $\mathbf{u} = (a, b)$, $\mathbf{v} = (1, 1)$, squared, then divide by 4.
- **#28** ($\langle f, g \rangle$ on $C[0, 1]$). Multiply out $f(t)g(t)$ and integrate term by term from 0 to 1.

§4 Problem set

P1. Apply Gram–Schmidt to $\mathbf{x}_1 = (1, 2, 2)$, $\mathbf{x}_2 = (4, 1, 1)$, then give an orthonormal basis of $\operatorname{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$.

P2. Find an orthogonal basis of the subspace of $\mathbb{R}^4$ spanned by $(1, 1, 1, 1)$, $(1, 1, 1, 0)$, $(1, 1, 0, 0)$.

P3. Find the QR factorization of the 3×2 matrix A with columns $(1, 2, 2)$ and $(4, 1, 1)$.

P4. Find the least-squares solution and the least-squares error of $A\mathbf{x} = \mathbf{b}$ for A with rows $(1, 0)$, $(0, 1)$, $(1, 1)$ and $\mathbf{b} = (1, 1, 0)$.

P5. With A from P3 and $\mathbf{b} = (1, 0, 0)$, find the least-squares solution using the QR factorization, and check it with the normal equations.

P6. Suppose the columns of A are orthonormal. Show that the least-squares solution of $A\mathbf{x} = \mathbf{b}$ is $\hat{\mathbf{x}} = A^{\mathsf{T}}\mathbf{b}$, and that $\operatorname{proj}_{\operatorname{Col} A} \mathbf{b} = AA^{\mathsf{T}}\mathbf{b}$.

P7. On $\mathbb{P}_2$ use the inner product given by evaluation at 0, 1, 2. Compute $\langle p, q \rangle$, $\|p\|$ and $\|q\|$ for $p = 1 + t$ and $q = t^2$. Then apply Gram–Schmidt to $1, t, t^2$.

P8. Use the Cauchy–Schwarz inequality to show:

(a) $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ for real numbers a, b, c;

(b) for every $f \in C[0,1]$,

$$\Big(\int_0^1 f(t)\,dt\Big)^2 \le \int_0^1 f(t)^2\,dt.$$